MOOD DETECTION SYSTEM

Software Requirements Specification (SRS)

Project Name: Mood Detection System

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1. Introduction

1.1 Purpose

The **Mood Detection System** is designed to analyze human emotions from text data. Users can either input text manually or upload a chat file (e.g., WhatsApp conversation). The system classifies messages into **Positive, Negative, or Neutral** categories, providing both message-level and overall mood analysis. The system also supports data visualization and CSV export.

1.2 Scope

- Detects sentiment in individual messages.
- Handles both small text inputs and large chat files efficiently.
- Generates tabular and graphical representations of sentiment results.
- Exports analysis to CSV for further use.
- Built in Python using NLP tools such as TextBlob.

The system is intended for:

- Personal use (to analyze one's chat moods).
- Researchers studying text-based emotional patterns.

- Social media or chat analysis tools integration.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
NLP	Natural Language Processing, a branch of AI for text analysis.
CSV	Comma-Separated Values, a file format for structured data.
GUI	Graphical User Interface, the visual interface for users.

1.4 References

- TextBlob Python library: https://textblob.readthedocs.io
- Python 3.x documentation: https://docs.python.org/3/
- IEEE SRS Standard (IEEE 830-1998)

2. Overall Description

2.1 Product Perspective

The Mood Detection System is a **standalone software application**. It can be extended later as a web or mobile application. It does not rely on external APIs for basic functionality but may require Python libraries like TextBlob, Pandas, and Matplotlib.

2.2 Product Functions

1. Accepts **manual text input** from users.
2. Allows **file upload** (e.g., WhatsApp chats in text format).
3. Processes input to classify messages as **Positive, Negative, or Neutral**.
4. Calculates **overall mood statistics**: counts and percentages of each category.
5. Generates **tabular output** displaying sentiment per message.
6. Generates **graphs/charts** showing overall sentiment distribution.
7. Saves results into **CSV files** for external analysis.

2.3 User Classes and Characteristics

- **General Users:** People who want to analyze their chats.
- **Researchers/Analysts:** Users who analyze sentiment trends in text data.
- Users need basic knowledge of uploading files or entering text. No programming skills required.

2.4 Operating Environment

- **Software Requirements:** Python 3.8 or higher
- **Libraries:** TextBlob, Pandas, Matplotlib

2.5 Design and Implementation Constraints

- Must handle large chat files efficiently (up to hundreds of thousands of messages).
- Must provide meaningful feedback if uploaded file format is incorrect.
- Graphical output should be clear and intuitive.

2.6 Assumptions and Dependencies

- Users provide chat files in readable text format.
- System depends on pre-trained sentiment models (TextBlob).
- Internet is required only for library installation; not for core functionality.

3. Specific Requirements

3.1 Functional Requirements

ID	Requirement Description
FR-1	The system shall accept manual text input from the user.
FR-2	The system shall accept chat file uploads.
FR-3	The system shall classify each message as Positive, Negative, or Neutral.
FR-4	The system shall display message-level sentiment in a table.
FR-5	The system shall calculate and display overall mood statistics.

ID	Requirement Description
FR-6	The system shall generate graphical representation of sentiments.
FR-7	The system shall save results in a CSV file.
FR-8	The system shall handle large datasets efficiently.

3.2 External Interface Requirements

- **User Interface:** Text input box, file upload button, table for output, chart display, save button.
- **Hardware Interface:** Standard keyboard and display; no specialized hardware needed.
- **Software Interface:** Python libraries (TextBlob, Pandas, Matplotlib).
- **Communication Interface:** Not required for standalone version.

4. System Models (Optional)

- **Use Case Diagram:** User enters text or uploads chat → System processes messages → System outputs table, graphs, CSV.
- **Data Flow Diagram:** Input → Sentiment Analysis → Output (Table + Graph + CSV).

5. Appendices

- Sample input file formats.
- Installation instructions for Python and required libraries.